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Factors affecting participation in joint forest management in the West Bengal state of India

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ABSTRACT

This study analyses participation in joint forest management (JFM). The study is based on in-depth interviews with 150 households of Forest Protection Committees (FPC) in Paschim Medinipur district in the West Bengal state of India. Based on a Principal Component Analysis (PCA) of 14 participatory indicators, it follows that “social” aspects of participation are considerably less important in West Bengal than as found elsewhere in India, while the perception of the environment (first and third factor) stand out as the most important aspects of participation in JFM, jointly explaining almost half of the variance. Regression analyses indicate that household size, the number of forest committee meetings, religion of the household, willingness to pay of the household for the forest protection and the size of land holding are important variables for explaining variation in levels of participation. Results from estimating game theoretic models on participation among villagers in JFM indicates that in two out of the four possible situations, harmonious sharing of the benefits is an (Nash) equilibrium, but not unique. In all cases, institutional checks and balances are needed to guarantee mutual participation, which is possible only by strengthening the JFM management body.

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Introduction

Forest is an important source of livelihood for the forest (fringe) communities in India. The National Forest Policy (NFP) outlined and conveyed to State Governments a framework for creating massive people's movement through the involvement of village committees for the protection, regeneration and development of degraded forestlands situated in the vicinity of villages (MoEF, 1990). Joint forest management (JFM) is now the central policy of the forest sector in all the 28 states of India (Pangannavar, 2011). Considering the ruthless exploitation of forests in West Bengal, legislation on JFM involving local communities in the process of protection and management of the forest was introduced in 1989. The concept of JFM is based on care and share. Current participatory approaches, methodologies and tools are widely used in the forest sector (Warner, 2005; FAO, 2005; UN, 2005).

There is a wide literature supporting the case for JFM (Bhattacharyya et al., 2010; Das, 2011; Nayak and Berkes, 2008). The World Bank study reveals that the economic returns from JFM in West Bengal are considerable (Hill and Shields, 2008). Bandyopadhyay and Shyamsundar (2004) using the National Sample Survey data from 524 villages in five states in India finds that household participation in JFM has a significant positive impact on consumption. The findings of Ghate and Ghate (2010) based on the primary data collected from six villages in the state of Maharashtra in India indicate that the relationship of indigenous communities with the forest continues to be non-exploitative and non-commercial. In fact, it has to be recognised that non-timber forest products (NTFPs) extraction becomes an important incentive for forest preservation (Dutta et al., 2004).

In spite of the various advantages, JFM in India is beset with many problems. It has been alleged that the structure of JFM is found to be skewed towards the forest department (Tiwarei, 2005). It has been suggested that the present structure that is tilted towards the forest department, needs to be clearly balanced with equal opportunities and rights to the communities participating in JFM programmes (Upadhyay, 2003). The experience of West Bengal relating to the extent of women's involvement in JFM and their role in JFM suggests that despite women's greater involvement in JFM in NTFPs, they are commonly excluded from participating in community institutions for JFM. The state JFM resolution does not explicitly mention women as independent; nor does it mention a minimum representation by women (Sarker and Das, 2002; Sarker, 2009). Roy et al. (2001) observe that even when there is a lot of disagreement in a meeting and the meeting is inconclusive, the close interaction during the meeting may have positive long term implications. This is due to a close connection between meetings (frequency and attendance) and forest status through the resolution of internal conflicts, leading to the smoother operation of forest preservation activities. They stressed the need to involve all villagers (and especially women) in meetings.

West Bengal is one of the few states in India where the per capita forest area is rather low (0.01 ha) and the density of population around the fragmented forest is high. There is a tremendous pressure on the forest in the state because of socio-economic problems like poverty and unemployment in the forest (fringe) areas. Though problems have been tried to be tackled through formation of forest protection committees (FPCs), there are cases of illicit collection of timber and poaching of wildlife (GoWB, 2009). It has also been observed that there is a clear declining trend in biodiversity due to illegal logging and killing of animals in some areas of West Bengal (Tisdell et al., 2001). There is an urgent need for more cooperation to develop the forest through community participation as it is linked with the livelihood of the forest (fringe) community. As the pressure on land is already high in the state and agriculture is dominated by marginal farmers with low returns, sustainable JFM can generate livelihood support to the poor forest fringe people. Various studies show that in many institutional settings of rural India, forests are better managed when people's voluntary participation is secured (Ravindranath and Sudha, 2004; Lise, 2007).

The situation of voluntary participation by members of a JFM organisation can be analysed as a game theoretic model. A situation could be called harmonious if there are sufficient incentives for participation in JFM among JFM organisation members. If this is not the case, there could be various conflict situations such as the prisoners' dilemma and the chickens' game. An example of a harmonious JFM situation is the assurance game, where mutual participation has the highest preference. Different studies show that the behaviour and determinants of cooperation varies from region to region. There are a limited number of studies into the behaviour and attitude of the community in JFM. Moreover,

the Government of India enacted the “Forest Rights Acts” (FRA) in 2006, to reflect the importance of recognising forest dweller’s rights over forest land and NTFPs. It has been argued that these rights over resources and the responsibility to conserve resources can only be ensured through participatory JFM (Sathyapalan and Gopinath Reddy, 2010). JFM is still very much relevant and only a very small percentage of forest dependent people have been given rights under the FRA. It is of interest to analyse the JFM in the post-FRA regime, which is done in this paper.

The impact of JFM activities on forest conservation and economic upliftment of forest fringe dwellers vary widely even for the FPCs characterised by similar resource parameters. We need to investigate the main factors affecting participation in JFM. This paper explores the qualitative and quantitative link between participation and socio-economic variables to establish when people are most likely to participate in JFM. To establish that link, we have collected data from 150 households who are members of FPCs in the Paschim Medinipur district of West Bengal. The overall aim of this paper is to find drivers behind people’s participation in JFM. A Principal Component Analysis (PCA) has been used to find which factors enhance people’s participation in JFM, whereas regressions link socio-economic variables to variation in levels of participation. Additionally, this paper explores whether there are incentives to cooperate in JFM by estimating game theoretic models, hence to verify whether JFM is harmonious or not.

The outline of this paper is as follows. Materials and method section entails a description of the study area and highlights the characteristics of the collected primary data for 150 households. Results and discussion section analyses the primary data employing a PCA to find factors driving participation in JFM. The derived factors are then used in a multiple regression analysis to identify the variables affecting these factors. Furthermore, a game is formulated among people competing for using the forests adjacent to their villages. Rather than choosing one particular game, we have left the game of people’s participation in JFM as an open option. This game is estimated and the results are discussed. The final section concludes.

Materials and method

Description of the study area

Our study area is in the district of Paschim Medinipur in the state of West Bengal in India within the geographical coordinates of 21°36′22″57″N latitude and 86°33′88″11″E longitude. The district is one of the backward districts in West Bengal with 34% of families below the poverty line. In recent times there has been a huge political disturbance in the region concerning the issue of development. The territory is of undulating character with red and unproductive soil and the climate is basically hot and humid with a winter of short duration (GoWB, 2009). The forest type here is tropical dry deciduous. Forest management is guided by a coppice system with a short rotation period of 10 years. The total forest cover in the district is 21.2%, whereas the state average is 15.6%. The state of West Bengal played a pioneering role in JFM in India with its beginning in Arabari in Medinipur district in 1971. The majority of forests have created FPCs through JFM. There are four forest divisions in the district of Paschim Medinipur namely Rupnarainpur, Midnapore, Kharagpur and Jhargram. Until March 2008, the number of FPCs in Paschim Medinipur was 1299 which was 31% of the total number of FPCs in the state of West Bengal and the number of beneficiaries from FPCs in the district was 0.15 million. According to the resolution of Government of West Bengal in 2008, a family living in the vicinity of the forests shall have the option of becoming a member of the JFM committee, when such a family is interested in the work of forest protection. There shall be joint membership of each household (i.e. once the husband is a member, the wife automatically becomes a member too and vice versa). Each JFM committee has an Executive Committee (EC) to carry out various activities assigned to the committee. The EC consists of seven persons including three elected representatives of the beneficiaries subject to the condition that at least one member will be a woman and/or a tribal. The members of the JFM committee are eligible for collecting NTFPs like twigs, grass, fruits, flowers, mushrooms, etc., whereas members of the JFM committee receive 25% of the net sales from firewood and poles. The main functions of the JFM committee and EC include ensuring protection of forest and wildlife, preventing trespasses,

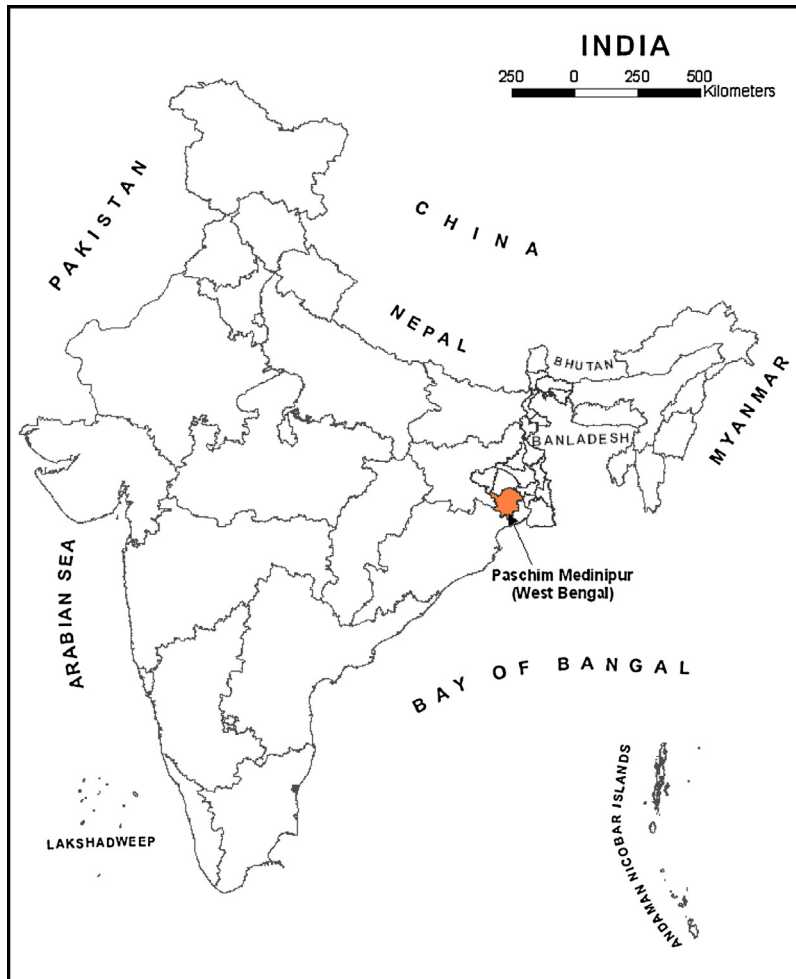


Fig. 1. Location of the study area.

encroachment, free grazing, fires, poaching, theft and damage, and ensuring smooth harvest of the forest produce by the Forest Department (GoWB, 2008).

Since the main aim of the paper is to find the attitude of the forest community towards JFM, we designed the questionnaire accordingly. The sample consists of 150 households selected from 31 FPCs. We have used stratified random sampling for the selection of FPCs. We first selected the ranges on the basis of preliminary interaction with the District Forest Officers (DFO) of Medinipur. After the selection of ranges, visits were made to these ranges for selection of FPCs and households which were chosen randomly. The FPCs were scattered over six forest range offices, namely Garhbeta, Hijli, Bhadutala, Humagrh, Amlagarh and C. K. Road. The data has been collected during the period August 2010 until January 2011. The sample of FPCs was evenly distributed to capture the variation in forest conditions. The average area of the FPCs under study is calculated as 175 ha and the average number of families per FPC is calculated as 133. The FPCs were formed between 1991 and 2001. The average forest composition of the surveyed FPCs is: natural forests—66%, plantation—18% and barren—8%. The location of the study area is presented in Fig. 1.

Description of the data collected

In total 150 household heads have been interviewed with the help of a structured questionnaire designed for primary data collection. The questionnaire included different household characteristics like socio-economic characteristics (age, education level, religion, caste, etc.), attitude of the households towards JFM, participation in JFM, and benefits from JFM, etc. In the end, 25 quantitative variables have been derived for the present study. Some of these variables are qualitative in nature and a psychological response score has been taken on 1 – 5 scale to convert them into quantitative variables, where 1 represents “totally disagree”, 2 “disagree”, 3 “neutral”, 4 “agree” and 5 “totally agree”. Capital of the household consists of power tiller, tube well, spray machine, thrasher, mobile, gas oven, pumpset, power tiller, etc. We have derived the value of capital by valuing these assets at current market prices. The payoff is calculated as the imputed income from NTFPs. The main items of NTFP include fuel (like firewood, dry leaves), food (like fruits, mushroom, potatoes, tubers and honey), medicinal, ritual and ornamental forest products (flowers, leafs etc.). Leaves of the sal tree are used to make indigenous plates. The families also collect fodder for the livestock which numbers about five per family. We have found more than 50 products collected from forest. NTFPs were valued at market prices to get the payoff. Forest Dependence (FORDEP) was calculated as the ratio of payoff over the total income of the household. Per capita consumption (CONPC) of the household was calculated as the ratio of reported annual consumption expenditure and the family size. The average value of PAYOFF and FORDEP is found as respectively Rs. 20,518 and 0.41 (Table 1).

The variables in Table 1 can be divided into the following categories:

- #1 – 14: Variables describing the perception of the interviewed persons concerning the village meetings and the state of the forest. These variables will be interpreted as indicators of people's participation in JFM in the sense that they reveal their personal opinion on the state of the forest in the village, which is a result of the quality of forests. These will be aggregates (see below) into the main dependent variables.
- #15–24: Independent variables that will explain the variation in the derived indicators of participation in JFM (the hypothesised and expected sign in the regression analysis is shown in the brackets).
 - #15–16: Characteristics of the interviewee. In general, more education and being female would translate into a higher level of participation (positive sign).
 - #17: The impact of family size on level of participation is more difficult to establish, but smaller families could be argued to be more participatory, because it will be easier to agree on a participatory strategy in a smaller group (Bardhan, 2000) (negative sign)
 - #18: The impact of religion on participation is not a priori clear, even though being from the majority religion (Hindu) may lead to more participation (positive sign).
 - #19: the number of meetings is a clear indication of the level of activity related to JFM. With more meetings the level of participation is expected to be higher (positive sign).
 - #22: also forest dependence is known to be strongly linked to the level of participation (Lise, 2007) (positive sign).
 - #20 – 21, 23 – 24: economic variables representing various aspects that could influence the budget of the family. A higher WTP for JFM would either indicate that the forest is important or that the interviewee has a high income. If the forest is important a higher participation could be expected, but a higher income does not necessarily lead to higher participation. The level of household consumption, household capital and land ownership are an indication of income level, but there is a priori no clear link to participation.

To evaluate all variables simultaneously and, thus, to characterise the role of people in relation to forests, leading to the key factors affecting people's participation, a PCA is used (Harman, 1967). Furthermore, the derived *factors*, which are a proxy for people's participation in JFM, are used in a multiple regression analysis (Ordinary Least Squares) to explain variation in people's participation

Table 1

Names labels, and units of the quantitative variables.

No.	Relevant question asked	Label	Unit	Min	Max	Mean
1	Are you undertaking tree planting work?	PLANT	–	1	5	3.62
2	Are you contributing money for JFM development?	MONEY	–	1	5	1.82
3	Are you benefiting from JFM?	BENEF	–	1	5	3.55
4	What is the frequency of the meetings of Managing Committee?	FREQU	–	1	5	3.31
5	To which extent can you agree with the discussions in the managing committee?	AGREE	–	1	5	3.09
6	To which extent can you influence the decisions of the Managing Committee?	INFLU	–	1	5	2.97
7	To which extent are the discussions of the meeting relevant?	DISCU	–	2	5	3.49
8	To which extent do you gain from such meetings?	GAINM	–	1	5	3.57
9	To which extent do you make suggestions in the meetings?	SUGGE	–	1	5	2.58
10	To which extent are you interest in the re-generation of forest?	REGEN	–	1	5	3.71
11	To which extent do you consider the current state of forest degraded?	DEGRA	–	1	5	3.85
12	Has there been satisfactory improvement in the conditions of the forest because of JFM?	CONDI	–	1	5	3.82
13	Has JFM led to satisfactory improvement in the socio-economic level of villagers?	SOCIO	–	1	5	2.81
14	What is the quality/quantity of resources like fodder and fuel wood extracted from the forest?	QUALI	–	2	5	3.52
15	How many years of education do you have?	EDUC	Years	0	15	4.48
16	What is your gender?	WOMEN	1 if true	0	1	0.04
17	What is the size of your household?	HHSIZE	#	1	12	5.22
18	Are you a Hindu?	HINDU	1 if true	0	1	0.57
19	What is the number of meetings held last year?	MEETN	#	0	12	3.29
20	What is your willingness to pay for the forest?	WTPFO	Rs	0	15000	351
21	How much land do you own?	LAND	Decimal	0	1300	130
22	Calculated variable: household PAYOFF divided by INCOME	FORDEP	–	0.01	1.00	0.41
23	Calculated variable: household consumption divided by HHSIZE	CONPC	Rs	223	5874	979
24	Calculated variable: household capital divided by HHSIZE	CAPPC	Rs	0	18183	3721
25	Calculated variable: imputed income derived from the forest	PAYOFF	Rs	1009	92331	20518

in JFM.¹ The regression analysis can only establish a link between socio-economic variables and one participation variable at the time, but cannot analyse the interlinkages among participation variables. To study these interlinkages, a non-cooperative game model is formulated and estimated to analyse whether there are incentives for voluntary people's participation in JFM. This can be used to verify

¹ In applying these statistical techniques, Version 12.0 of the Statistical Package for Social Science (SPSS) is used.

Table 2

Personal features.

Features	Frequency	%
<i>Age groups</i>		
20 – 30	4	3
31 – 40	28	19
41 – 50	53	35
51 – 60	41	27
60+	24	16
<i>Education levels</i>		
1 Illiterate	15	10
2 Primary school	9	6
3 Secondary school	67	45
4 High school	4	3
5 University (two years)	5	3
<i>Religion</i>		
1 Hindu	89	59
2 Muslim	5	3
3 Sarna	56	37
<i>Caste</i>		
1 General	57	38
2 Scheduled caste (SC)	18	12
3 Scheduled tribe (ST)	60	40
4 Other backward caste (OBC)	15	10

whether JFM is in harmony or in conflict in the studied region in India and under which conditions either conflicts emerge or harmony prevails.

Characteristics of the collected data

In order to obtain an insight into the composition of the interviewed 150 people, Table 2 shows the distribution of interviewees according to personal features.

Table 2 reveals that about half of respondents are of the age between 30 and 50. The majority of the household heads have reached secondary school with 10% of the household heads being illiterate. Sarna is the religion of most of the tribal households (37%) and 40% of the households belongs to the Scheduled Tribe caste. The forest area per family is calculated as 1.32 ha per family for the sample FPCs.

In Table 3 we have grouped interviewees with respect to forest area per family in ha. It turns out that 41% of the interviewees lie below 1 ha of forest land while 29% of the interviewees lie above 2 ha of forest area. The forest area per family for the sample FPCs is calculated as 1.06 ha while the official figure for state average of the same is about 18 ha. The names of the villages from which the samples have been drawn are also mentioned in the last column of Table 3.

Results and discussion

Principal component analysis of people's participation

To determine the most important factors affecting people's participation in JFM, a PCA is undertaken on 14 participatory indicators (the first 14 variables in Table 1).² These 14 variables have been chosen as they represent different aspects of participation, namely participation in the JFM meetings, contribution to the forest, benefits from the forest and perception of the forest quality. Table 4 shows the result. The first four factors turn out to have eigenvalues with a value greater than one, leading to four factors.

² See also Lise (2007).

Table 3

Grouping participation according to forest area per family.

Participation group	Forest area per family (ha)	Number of interviewees	% of interviewees	No. of villages	Name of villages
Group 1	Below 1	61	41	14	Upar Sole, Chandabila layakpara, Dhadika Mahila, Balijuri, Ekaria Fulmonipur, Dangarpara, Barguitra-Dhanghori, Nohari, Rajbandh-Sarbat, Nalbona, Dakhinshole, Dhobaberia-Jemuasole, Ajodhagarh, Paschim Pathri
Group 2	1 – 1.99	50	33	10	Raghunathpur, Amkopa, Lachangarh, Nachna Ajodhyagarh, Pathri Sole Namo, Birbandhi, Nachanjam, Shyampur, Tangasole, Ghatmura
Group 3	Above 2	39	26	8	Dhabani, Dhannyashole, Ichal Konda, Dhanayardihi Garki, Indkuri, Quelibandh, Banachapra, Peruajole

Table 4

PCA of 14 indicators of people's participation in JFM.

Description	Code	Forest condition	Importance of the meetings	Benefit from the forest	Benefit from the meetings
		Factor 1	Factor 2	Factor 3	Factor 4
Are you undertaking tree planting work?	PLANT	0.679	0.273	0.109	0.157
Are you contributing money for JFM development?	MONEY	0.210	0.748	0.209	–0.139
Are you benefiting from JFM?	BENEF	0.422	0.643	–0.152	0.055
What is the frequency of the meetings of Managing Committee?	FREQU	0.401	0.536	0.054	0.446
To which extent can you agree with the discussion of the Managing Committee?	AGREE	0.186	0.252	0.314	0.456
To which extent can you influence the decisions of the Managing Committee?	INFLU	–0.001	0.515	0.386	0.353
To which extent are the discussions of the meeting relevant?	DISCU	0.231	0.138	0.428	0.621
To which extent you gain from such meetings?	GAINM	0.211	0.030	–0.026	0.874
To which extent you make suggestion in the meeting?	SUGGE	–0.109	0.690	0.194	0.268
To which extent are you interested in the regeneration of forest?	REGEN	0.665	–0.040	0.290	0.359
To which extent do you consider the current state of forest degraded?	DEGRA	0.800	0.100	0.232	0.123
Has there been satisfactory improvement in the conditions of the forest because of JFM?	CONDI	0.821	0.114	0.270	0.143
JFM has led to satisfactory improvement in the socio-economic level of villagers?	SOCIO	0.310	0.197	0.790	0.120
What is the quality/quantity of resources like fodder and fuel wood extracted from the forest?	QUALI	0.288	0.097	0.832	0.118
Variance explained (%)		39.3	10.4	8.4	7.8

Note: Rotation method: varimax with Kaiser normalisation. Rotation converged in six iterations ($N = 150$).

Interpretation of the result in Table 4 yields that the most important component of people's participation is related to the *forest condition*, explaining 39% of the variance. There are four dominant indicators of participation (factor loading larger than 0.5 in absolute terms). In the first factor people participate in plantations, and are concerned about the level of regeneration, degradation and the condition of the forest.

The second factor is considerably less important than the first factor, explaining 10% of the variance, representing their economic gain and input into the meetings, which can be summarised as the *importance of the meetings*. There are five dominant indicators of participation here. In the second factor, people tend to gain in the form of money or benefits and contribute in the form of meeting frequently, their influence and ability to make suggestions during the meetings.

The *benefit from the forest* is the third factor, best captures the joined meaning of the dominating indicators in the third factor, explaining 8% of the variation. There are only two dominant indicators of participation in this factor. A high value in the third factor indicates people's need for an improvement of the socio-economic level of the villagers and a good quality of the products that can be collected from the forest.

Benefit from the meetings would best capture the meaning of the last factor, explaining 8% of the variance. The fourth factor consists of two participatory indicators, namely the extent to which discussions in the meetings are useful and whether the meetings are useful. Also the indicator 'agreement with meetings' has the largest factor loading here, but has a value just below 0.5, which makes it a weakly dominating indicator of participation.

These results can also be compared to Lise (2007), where largely the same questions were asked during a survey in three different states in India, namely Haryana, Jharkhand and Uttarakhand. However, this paper differs by including the perception of the environment to the total set of variables, which was not the case in Lise (2007). Nevertheless, the following similarities were found. The second factor is similar to "economic" which was also found to be the second factor in all three states. The fourth factor is similar to "social (internal)", which was the first factor in all the three states. Hence, "social" aspects of participation is considerably less important in West Bengal than as found in the previous study. The perception of the environment (first and third factor) stand out as the most important aspects of participation in JFM.

Multiple regression analysis of people's participation

To explain the driving force behind the four factors of people's participation in JFM, we have also undertaken a multiple regression analysis. The following model is estimated by using Ordinary Least Squares (OLS):

$$\text{Factor}_i = \text{constant} + \beta_1 \text{EDUC} + \beta_2 \text{WOMEN} + \beta_3 \text{HHSIZE} + \beta_4 \text{HINDU} + \beta_5 \text{MEETN} + \beta_6 \text{WTPFO} + \beta_7 \text{LAND} + \beta_8 \text{FOREDEP} + \beta_9 \text{CONPC} + \beta_{10} \text{CAPPC} + \text{error} \quad (1)$$

Eq. (1) shows that the four factors, which were found with the PCA, are used as the dependent variables. Ten variables enter the regression equation as descriptive variables.

Table 1 shows the meaning of the variables, which are included in the regression. The result is presented in Table 5.

Interpretation of Table 5 yields a number of additional insights into the process of people's participation in JFM. First of all, all included variables are at least once significant in explaining the variation in participation.

Secondly, as shown in the first regression, higher forest dependence goes hand in hand with a lower willingness to pay in explaining the forest condition. This shows that people who depend on the forest may not have the means to pay for its regeneration and therefore have a lower willingness to pay. The best chances of voluntary participation can be found among the villagers who depend highly on the forest and perceive the quality of the forest as good (Lise, 2007). Additional drivers for a good state of the forest are not being a Hindu, a high number of meetings and a high amount of land ownership.

Table 5
Regression analysis to explain variation in people's participation.

Description	Code	Forest condition		Importance of the meetings		Benefit from the forest		Benefit from the meetings	
		PC1		PC2		PC3		PC4	
(Constant)		−0.878**	(0.384)	−0.954**	(0.372)	−0.409	(0.423)	−0.989**	(0.420)
Years of education (10^{-2})	EDUC	−0.128	(1.822)	5.746***	(1.763)	0.282	(2.009)	−0.035	(1.993)
Gender	WOMEN	0.536	(0.384)	0.597	(0.372)	−0.513	(0.423)	1.130***	(0.420)
Household size	HHSIZE	0.032	(0.045)	−0.080*	(0.044)	0.072	(0.050)	0.131***	(0.050)
Religion	HINDU	−0.293*	(0.165)	0.281*	(0.160)	0.094	(0.182)	0.402**	(0.181)
Number of meetings	MEETN	0.094*	(0.055)	0.220***	(0.053)	−0.021	(0.060)	−0.028	(0.060)
Willingness to pay for the forest (10^{-4})	WTPFO	−1.320***	(0.433)	0.835**	(0.419)	−0.330	(0.478)	0.518	(0.474)
Land ownership (10^{-3})	LAND	1.340**	(0.658)	1.430**	(0.637)	−1.340*	(0.725)	−0.958	(0.720)
Forest dependence	FORDEP	0.972***	(0.277)	0.000	(0.268)	0.360	(0.305)	0.358	(0.303)
Consumption per capita (10^{-4})	CONPC	1.310	(1.320)	0.484	(1.280)	−2.930**	(1.450)	0.847	(1.440)
Capital per capita (10^{-4})	CAPPC	−0.272	(0.275)	−0.156	(0.267)	1.030***	(0.304)	0.004	(0.301)
R ² adjusted		0.2245		0.2741		0.0578		0.0728	

Note: The presented values of five variables (indicated with 10^{-x}) need to be divided by 10^x . The value in the brackets denotes the Standard Error.

* Significant at the 0.10 level ($P < 0.10$).

** Significant at the 0.05 level ($P < 0.05$).

*** Significant at the 0.01 level ($P < 0.01$).

Table 6
Payoff matrix for the participation game.

		Villager 2: (contender)	
		Participate	Do not participate
Villager 1: (challenger)	Participate	x, x	b, a
	Do not participate	a, b	y, y

Thirdly, according to the second factor, there are six drivers for evaluating the importance of the meetings. The six drivers are being well-educated, having a small household, being a Hindu, a high number of meetings, a high willingness to pay and a high amount of land ownership. Hence, relatively rich people with good education and being a Hindu tend to benefit from the forest and give importance to the village meetings on JFM.

Fourthly, there are three drivers for deriving benefits from the forest, namely a low level of land ownership, a low consumption per capita, and a high level of possessions in the form of capital per capita. This shows that different ways of measuring wealth are the drivers behind deriving benefits from the forest. This result reveals that people with low levels of land ownership and low per capita consumption, but with a high capital per capita in non-land goods tend to benefit from the forests.

Fifthly, according the fourth factor, there are three drivers for evaluating the benefit from the meetings. These three drivers are all related to characteristics of the person interviewed, namely whether the person is a woman, has a large household and is a Hindu. A female Hindu with a large family will benefit the most from the forest.

Finally, we discuss the derived signs of the statistically significant estimates. Most estimates have the hypothesised and expected signs. However, there are some exceptions. Larger households benefit more from the forests. This is possible as these families have more man-power to collect forest products and will therefore perceive their benefits more positively. Persons who are not a Hindu perceive the condition of the forest as better. This is a personal taste and as such this result is not surprising and also does not contradict with our expectation (see Description of the study area section). Last of all the estimates related to wealth are mixed. This is also in line with the expectation.

The participation game among people in JFM in India

For studying the opportunities of local people to voluntarily participate in the management of forests adjacent to their villages, we propose a general non-cooperative game model, without specifying this game beforehand. The strategy is to choose the level of participation in JFM. Here participation measures how a villager perceives the local organisation in the village to manage an adjacent forest. A participating villager adheres to rules and codes of conduct that are prevalent in the village meetings for JFM. This participation is awarded with the right of access to the forest from which they can reap benefits (=their net payoff). We refer to this situation as the *participation game*.

In order to formalise possible conflicts, which can emerge between villagers, we will focus on the case with n villagers contesting for access to the forest. Then we can distinguish between villager 1, the challenger, and villager 2, the contender, which is composed of all other villagers contesting for the same forest. For that we need to assume that the challenger interprets the actions of other villagers as a simultaneous move. Hence, we are dealing with a 1 versus $n - 1$ person's game (see also Lise, 2007).

The simplest form of such a game consists of only two persons, who have a choice between two alternatives: to participate or not. When both villagers participate they obtain x . When one villager participates, while the other does not, the single participant keeps the rules as formulated during the village meetings, obtaining b , while the other does not follow the set rules, reducing the protection of the forest, obtaining a . This 'cheating' can be detected and deterred through 'social fencing', where the rule-abiding villager spots cheating. This can be reported to the forest guard. Finally, when both deviate, rules are not adhered to by both villagers, obtaining y . Table 6 shows the resulting payoff matrix.

A possible outcome of this game is a prisoner's dilemma, where the dominating strategy is to not participate and deplete the forest, while it would lead to collective better results when both villagers

Table 7
Assigning people's level of participation.

Level of participation of challenger	Level of participation of contender	Payoff group
'Participate'	'Participate'	X
'Participate'	'Do not participate'	B
'Do not participate'	'Participate'	A
'Do not participate'	'Do not participate'	Y

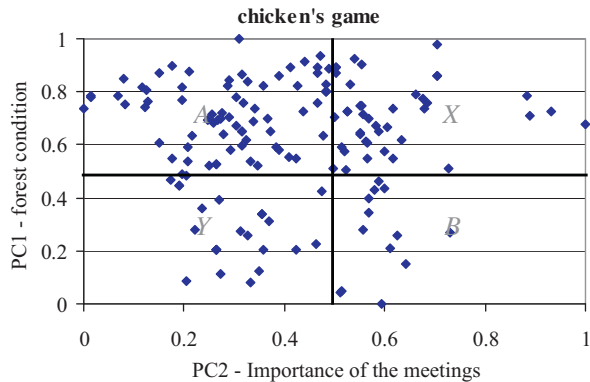


Fig. 2. Scatter plot of importance of the meetings with forest condition.

would participate, keeping the regeneration rate of the forest optimal. Hence the following inequality could hold:

$$a > x > y > b \tag{2}$$

It is also possible that there are other types of games.

For estimating the participation game we need to quantify the payoff which is measured as the benefits from forest resources (the variable PAYOFF as shown in Table 1).

Next, the strategy of the challengers is the strategy of the villagers and is measured as the level of participation (chosen as the second or fourth factor as derived in Principal component analysis of people's participation section). The strategy of the contenders is the strategy of all other villagers as perceived by the challenger, can be derived by taking the perception of forestry attributes (chosen as the first and the third factor, as derived in Principal component analysis of people's participation section). The first and the third factor is in a way the perception of the villagers of the joint action of all other villagers. Hence, the questions leading to variables #1 – 14 are based on Lise (2007) and adapted to the local situation. In contrary to Lise (2007) the indicators of participation (variables #1 – 9,13) and the state of the forest (variables #10 – 12,14) are pooled together. In this way a direct link can be made to the game theoretic model. The factors with the more personal preferences can be identified as the choice of the challenger, whereas factors that represent the general state of the forest will be identified as the choice of the contender, namely the collective choice of $n - 1$ other villagers.

To interpret the value of the strategy, of the challenger and the contender it is converted into a fraction between 0 and 1.

It is possible to assign the payoffs into four payoff groups by taking 0.5 as the threshold value. We define participation values above 0.5 as participative behaviour in the sense that a villager has a participatory attitude, while a participation value 0.5 indicates that a villager is not participating. This simple way of splitting the payoffs is referred to as the Simple Threshold Method (STM) (Lise et al., 2005). Assigning the payoffs is done as shown in Table 7.

The STM as explained above is applied to derive the participation games. In order to show the concentration of participation choices, the choices of the challenger and the contender are plotted in Figs. 2–5. The choice of the challenger represents respectively the importance of the meetings (factor

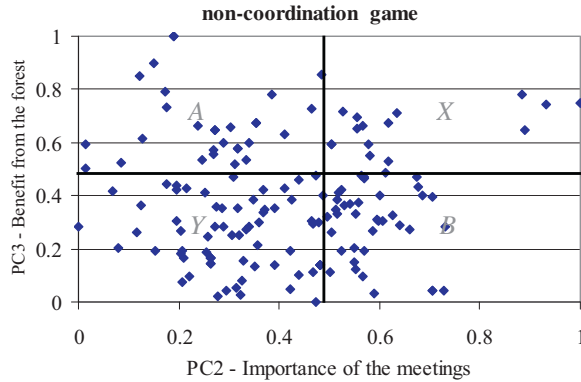


Fig. 3. Scatter plot of importance of the meetings with benefit from the forest.

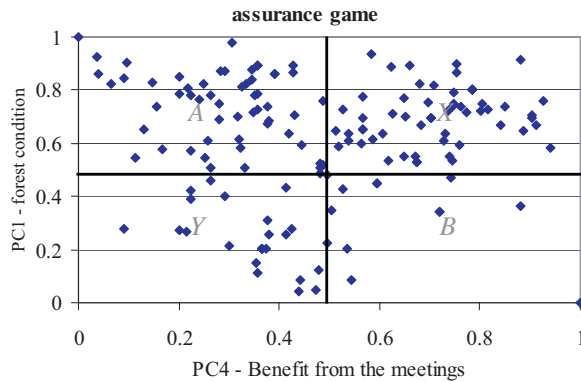


Fig. 4. Scatter plot of benefit from the meetings with forest condition.

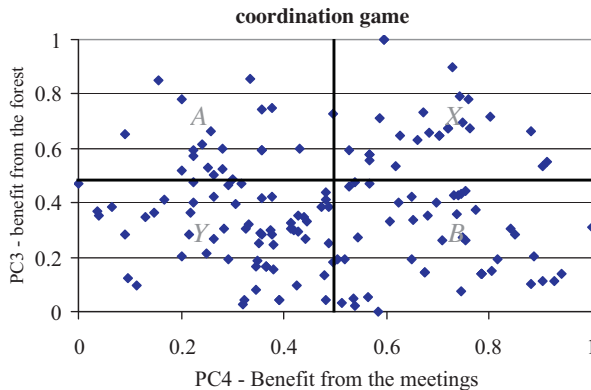


Fig. 5. Scatter plot of benefit from the meetings with benefit from the forest.

2) and the benefit from the meetings (factor 4). The choice of the contender represents the perception of the forest condition (factor 1) and the benefit from the forest (factor 3). Figs. 2–4 show the result for the Simple Threshold Method where a division into four payoff groups is indicated by the thick lines.

Table 8

Estimated participation games.

	<i>a</i>	<i>b</i>	<i>x</i>	<i>y</i>	Payoff order	Name of the game
Importance of the meetings with forest condition	22,626 (73)	14,590 (13)	22,393 (40)	14,192 (24)	$a > x > b > y$	Chicken's game
Importance of the meetings with benefit from the forest	25,012 (29)	20,751 (38)	19,792 (15)	18,631 (68)	$a > b > x > y$	Non-coordination game
Benefit from the meetings with forest condition	22,151 (55)	13,417 (11)	22,916 (58)	14,719 (26)	$x > a > y > b$	Assurance game
Benefit from the meetings with benefit from the forest	19,355 (19)	18,687 (44)	26,179 (25)	19,891 (62)	$x > y > a > b$	Coordination game

Interpretation of Figs. 2–5 already leads to an interesting outcome, namely that the most frequent occurrence of mutual participation (=X) is found with respect to benefit from the meetings and forest condition. This is shown in the figures by the concentration of data at the upper-right cell (=X) in Fig. 4. B is the right lower cell, A is the left upper cell and Y is the left lower cell (this is also indicated in the figures by putting an upper-case letters in the four cells).

The game estimation results, using the primary data as explained in Materials and method section, are presented in Table 8.

In the chicken's game and non-coordination game it is optimal for the players to take opposite actions (Lise, 2007). A chicken's game for the game on the importance of the meetings with the forest condition implies that the optimal strategy of the challenger is to choose the opposite of the strategy of the contender. In co-ordination and assurance game it is a matter of coordination to arrive at the equilibrium of the game even though mutual non-cooperation is also a Nash equilibrium. Hence, in none of the four games a clear-cut situation of mutual participation comes forward as the unique (Nash) equilibrium. This shows the fragility of JFM in studied area. Additional rules will need to be enforced to make JFM a success.

Conclusions

The study has identified factors determining people's participation in JFM. Statistical methods have been employed to evaluate the primary data obtained from 150 interviewees. The four most important factors that have emerged from the analysis are: (i) forest condition, (ii) importance of the meetings, (iii) benefit from the forest, and (iv) benefit from the meetings. These four factors explain 66% of people's participation in JFM and should be taken into consideration to enhance participation of people in JFM in the region of Paschim Medinipur district in India.

Unlike the study done by Lise (2007) in three different states in India, namely Haryana, Jharkhand and Uttarakhand, the "social" aspects of participation is considerably less important in West Bengal, while the perception of the environment (first and third factor) stand out as the most important aspects of participation in JFM.

A regression analysis has been undertaken to analyse the process of people's participation in JFM. Most of the estimated significant variables have the expected sign, whereas the exceptions can be explained by getting deeper into the context. From this we can draw the following lessons. Drivers for a good state of the forest are a high forest dependence (in line with Lise (2007)), not being a Hindu, a high number of meetings and a high amount of land ownership. Next, relatively rich people with good education and being a Hindu tend to benefit from the forest and give importance to the village meetings on JFM. Furthermore, people with low levels of land ownership and low per capita consumption, but with a high capital per capita tend to benefit from the forests. Finally, a female Hindu with a large family will benefit the most from the forest.

The issue of participation has been further explored. Participation in JFM is a multidimensional issue and therefore the estimated game consists of a decision on four alternatives as derived from PCA, in which FPC members can participate or not. The game analysis reveals that there are conflicts concerning sharing the forest benefits in the study area. To overcome these conflicts, JFM organisations

will need to be strengthened. There is scope for improving importance of the meetings and benefits from meeting to stimulate participation in JFM and achieve sustainable JFM in the study area in India.

Forests still play a great role in the livelihood of rural households in forest fringe areas. It is interesting to note that even though they depend heavily on the forest resources, many of the households do not agree with the view that JFM has led to the satisfactory improvement in forest conditions or socio-economic level of the villagers. The positive relation of the forest condition with the number of meetings implies that participation and exchange of opinions may enhance the forest condition.

Our results reveal high forest dependence of the forest fringe people. Forest conditions may be improved by lowering the forest dependence. While direct benefits flow to the members of FPCs from forest resources by way of sharing of usufructs; employment generation is generally rather limited. There is scope for improving traditional activities and allied land-based or skill-based activities like: (i) Agricultural development, (ii) promotion of small scale cottage industries based on NTFPs, (iii) agro-forestry and farm forestry practices including intercropping in plantations raised by the Forest Department, (iv) development of marketing facilities of NTFP and forest based products and (v) value addition of NTFP resources.

In India about 85,000 Forest protection communities are entrusted with the protection and management of nearby forests and more than 17 million hectares of degraded land are managed under JFM. JFM has certainly played a tremendous role in arresting the degradation of forests in India. But the performance of FPCs are not satisfactory everywhere. It has been found that community forest management is more successful than JFM. It is a possible future area of research to find the types of institutional changes needed to make the JFM programme sustainable.

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